Weekly Project Report – 4

Ahmedabad University

School of Engineering and Applied Sciences

Automated Multi-Parametric MRI Segmentation of Post-Treatment Glioma Sub-Regions

Using Classical Machine Learning and Contour-Based Methods

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***Abstract*—The current report presents the week 4 progress on automated segmentation of brain tumors with XGBoost on multi-parametric Brats MRI data. The 53 feature pipeline was established on the pipeline generation of voxel based on the construction of pipeline on the 4 MRI modalities (T1, T1ce, T2, FLAIR) of tumor/background dichotomy. It was maintained in dissimilar percentage of tumor to background 1:46 (in the view of the natural disequilibrium of the classes) in the form of v7 and 1:10 in the form of v8 since 136 test people had reached a Mean Dice of 0.7508 and a Mean IoU of 0.6269. The bugs which existed in the first codebase could be adopted in both of the versions and were identified and addressed also. It is also referred to as per-patient analysis and segmentation visualisation.**

I. Introduction

Furthering the work we did in week 3 which was implementing the pipeline using Random Forest with MRF refinement, we implemented gradient boosted tree approach using XGBoost [2]. XGBoost will be faster as it has `scale_pos_weight`, and applicable in solving class imbalance. It will also consume less model space and can be used in an environment that has a limit of only CPU, low-memorial space. It is a non-deep-learning binary (tumor vs background) classifier of 53 handcrafted features based upon four MRI modalities and on a per-voxel basis.

II. Feature Extraction and Pipeline

A. MRI Modalities and Brain Masking

Four co-registered 3D MRI volumes per patient are used: T1, T1ce (contrast-enhanced), T2, and FLAIR. All the brain modalities calculates a brain mask over non-zero voxels thus restricting features extraction and classification to brain tissue.

B. 53-Feature Vector

For each brain voxel, 53 features are extracted: 44 per-modality features (11 per modality $\times$ 4 modalities) plus 9 global features.

Per-Modality Features (11 $\times$ 4 = 44):

- **Raw intensity (1):** The normalized value of the intensity of the voxel of that modality.
- **Local mean – 3 scales (3):** Average intensity in $3\times3\times3$, $5\times5\times5$, and $9\times9\times9$ neighborhoods. The severity of the tumors is a scale-based characteristic depending on the locality of the tumor.
- **Local std – 3 scales (3):** Standard deviation in $3\times3\times3$, $5\times5\times5$, and $9\times9\times9$ neighborhoods. Local variation of tumors is enormous particularly at edges.
- **Gradient magnitude (1):** Computed using Sobel filters along all three axes as $\sqrt{G_x^2 + G_y^2 + G_z^2}$, capturing edges and tumor boundaries.
- **Gaussian blur – 2 scales (2):** The multi-scale soft context power is at σ=1 and σ=2.
- **Difference of Gaussians (1):** This refers to the difference of the two blurred images using Gaussians which is a band-pass filter in order to possess structures of middle scale.

Global Features (9):

- **Spatial coordinates (3):** Normalized (x, y, z) voxel position within the volume, encoding location context in the anatomy. The sub-regions of the tumor are selectively implanted in selected places of the brain therefore, the spatial location is weak though advantageous anatomical antecedent.
- **Cross-modality ratios and differences (6):** One ratio or difference is computed for each of the $\binom{4}{2} = 6$ unique modality pairs: (T1, T1ce), (T1, T2), (T1, FLAIR), (T1ce, T2), (T1ce, FLAIR), and (T2, FLAIR). These capture relations unavailable in one modality including, a T1ce/T2 ratio elicits into the limelight the augmenting tumor core, that is bright in T1ce and dim in T2 and T1ce-FLAIR can distinguish between augmenting tumor core and the surrounding edema.

C. Voxel Sampling Strategy

No training of voxel can be mathematically feasible. Two sampling plans that were attempted were:

- **1:46 ratio (v7)** – trading off natural class frequency (≈46 background per tumor voxel). `scale_pos_weight = 46.77`.
- **1:10 ratio (v8)** – downsizing a background (10 background per tumor voxel) as much as possible to enhance the effective tumor fraction.

III. Bug Analysis and Corrected Scripts

The original codebase was found to contain seven bugs in the codebase. Table I lists the critical ones and their relevance.

TABLE I
BUGS IDENTIFIED AND CORRECTED

#	Errors	Effect
1	Air/skull voxels included in background sampling	Wrong class ratio computed
2	Header says threshold 0.5; code uses 0.6	Misleading documentation
3	No `early_stopping_rounds`	Potential overtraining
4	`joblib.save` outside `__main__`	Crashes on import

Key Insight – The Accidental Bug: Bug #1 is notable because it accidentally *improved* performance. The intended 1:46 ratio was supposed to yield ≈2.1% tumor voxels in training, but due to the bug (air/skull counted in background), the effective ratio became ≈9.75% tumor – a more balanced distribution that likely boosted Dice from an estimated 0.65–0.68 to 0.75+.

IV. Hyperparameter Comparison: v7 vs v8

TABLE II
HYPERPARAMETER DIFFERENCES BETWEEN V7 AND V8

Parameter	v7 (1:46)	v8 (1:10)
TUMOR_BG_RATIO	46	10
MAX_DEPTH	8	5
learning_rate	0.1	0.05
subsample	0.8	0.6
min_child_weight	5	20
reg_alpha	–	0.1
reg_lambda	–	5.0
early_stopping_rounds	–	20

V. Results

A. Training Log (v7 – 1:46 Ratio)

The sample size used was 2,068, 069 with 53 features on which the training was conducted. The XGBoost logloss was also decreasing with the minimum of 0.623 at round 0 and 0.062 at round 299. The validation measures of the held out set were:

- Dice: 0.6322 IoU: 0.4623 AUC-ROC: 0.9897
- Background precision/recall: 1.00 / 0.98
- Tumor recall: 0.90 (high sensitivity)

B. Test Set Results – 136 Patients

Full-volume inference was run on 136 test patients at threshold = 0.6.

TABLE III
TEST RESULTS: V7 (1:46) VS V8 (1:10)

Metric	v7 (1:46)	v8 (1:10)
Mean Dice	0.7508	0.6282
Std Dice	0.1764	0.2041
Median Dice	0.8046	0.6701
Min Dice	0.0000	0.0000
Max Dice	0.9452	0.9383
Dice $\geq$ 0.9	104/136	2/136
Dice $\geq$ 0.7	104/136	61/136
Dice $\geq$ 0.5	125/136	104/136
Mean IoU	0.6269	0.4874
Model Size	1.7 MB	0.3 MB

Also good results demonstrate that v7 model is far better than that of v8 in all indices, the consequence of the natural imbalance of the classes `scale_pos_weight`) is better than the violent undersampling.

C. Train vs. Test Generalization

TABLE IV
TRAIN VS. TEST COMPARISON (OVERFITTING ANALYSIS)

Metric	Train	Test	Gap
v7 – 1:46 ratio			
Mean Dice	0.7494	0.7508	−0.0014
Median Dice	0.7943	0.8046	–
Mean IoU	0.6237	0.6269	–
v8 – 1:10 ratio			
Mean Dice	0.6216	0.6282	−0.0066
Std Dice	0.2016	0.2048	–

The performances of both models are very good with regard to generalization (train ≈ test). A little bit of underfitting (test > train) of the v8 model would not have been undesirable, and a stronger portion of regularization would have been beneficial and increasing `max_depth` or reducing `reg_lambda` would help.

D. Single-Patient Inference Example

For patient `BraTS-GLI-00517-100`:

- Volume shape: $182 \times 218 \times 182$ — Brain voxels: 1,698,590
- Feature matrix: $(1{,}698{,}590 \times 53)$, 360 MB
- Predicted tumor voxels: 181,668 (2.52% of brain)
- GT tumor voxels: 118,786
- **Dice: 0.7388 IoU: 0.5858**

VI. Segmentation Visualizations

Figures 1 and 2 represent the two qualitative outcomes of two validation patients. Each figure shows (left to right): the T2-FLAIR input slice, the expert ground truth mask (red), the model prediction (blue), and the overlay (green = overlap, red = missed, blue = false positive).

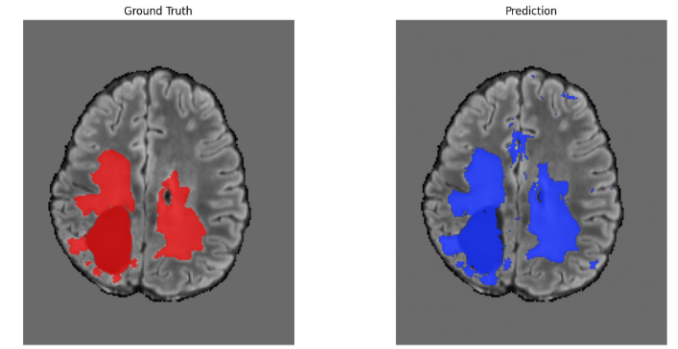

Fig. 1. BraTS-GLI-00517-100 — z=115 — Dice=0.739, IoU=0.586. The model captures the large tumor core well with some over-prediction into the surrounding edema.

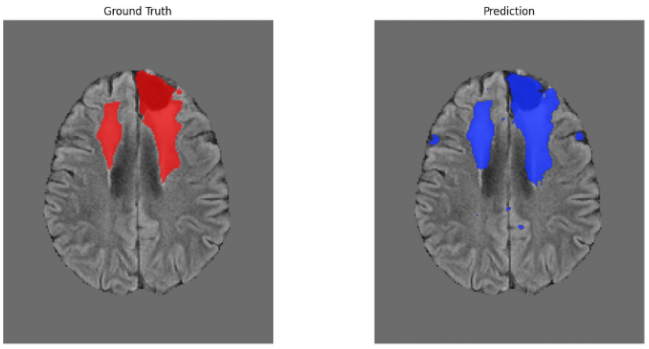

Fig. 2. BraTS-GLI-02092-105 — z=109 — Dice=0.731, IoU=0.575. Prediction correctly identifies the elongated tumor shape but includes some false positives in the contralateral hemisphere.

A. Best and Worst Case Patients

Figures 3 and 4 show the highest and lowest scoring patients from the v7 test set.

VII. Plan for Week 5

- **Pipeline 2 Integration:** Entire GLCM texture feature extraction pipeline and CRF based post processing pipeline according to Week 3.
- **Multi-label Segmentation:** Extend binary classification to three sub-regions (NCR/NET, Edema, Enhancing Tumor) using multi-class XGBoost.

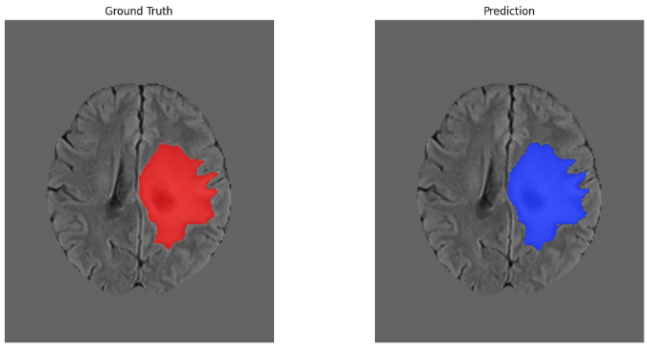

Fig. 3. Best performing patient – BraTS-GLI-02141-100 — Dice = 0.9452 (v7). Near-perfect overlap between prediction and ground truth.

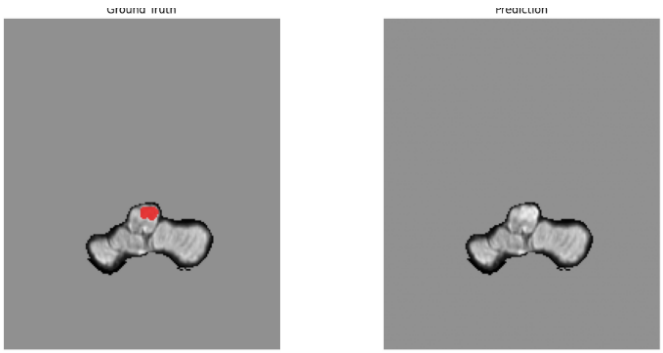

Fig. 4. Worst performing patient – BraTS-GLI-02767-100 — Dice = 0.0000 (v7). Complete segmentation failure – likely a very small or atypical tumor.

- **Threshold Optimization:** The largest Class-specific threshold optimization with precision and sub-region recall trade-off.
- **Error Analysis:** Find 11 patients of Dice < 0.5 to get familiar with failure modes.

VIII. Conclusion

The binary tumor segmentation pipeline using XGBoost from Week 4 was already fully functional. The performance of the model v7 (ratio 1:46) has a Mean Dice of 0.7508 and a Mean IoU of 0.6269 on 136 test patients. The model is very small at only 1.7MB with only CPU used. The analysis of the bug revealed that because of a mistake in sampling, a more balanced distribution was achieved in the end, resulting in a better performance than it would have achieved with a completely accurate model. The experiments done on model v7 and v8 in the form of controlled experiments validated the natural class distribution and strategy of correcting it using `scale_pos_weight` as the best approach. The model does not over/underfit because there is great generalization between train and test (gap < 0.002).

References

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[2] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," *Proc. ACM SIGKDD*, pp. 785–794, 2016.

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